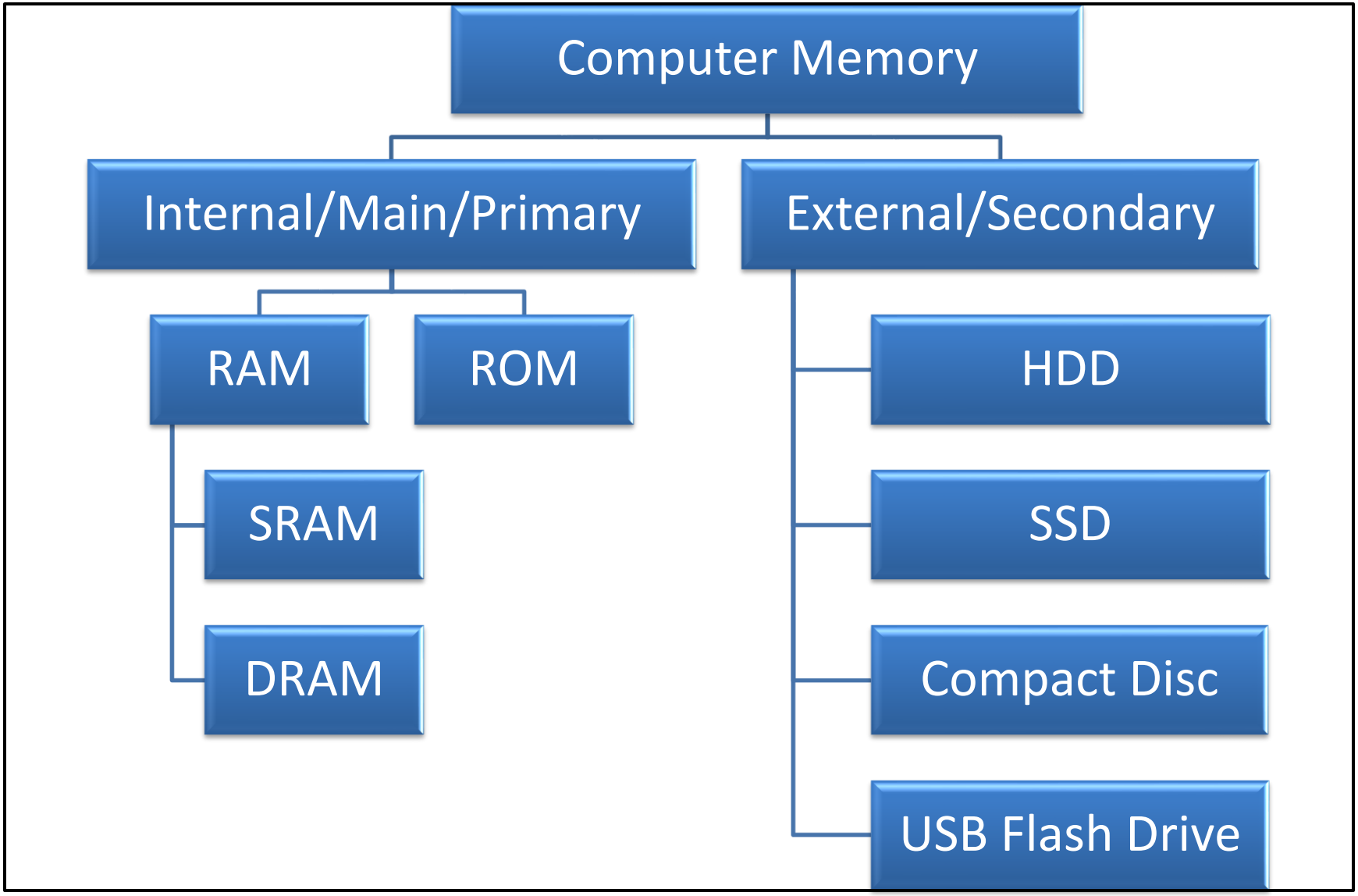
Applications of Information and Communication Technologies

Lecture-4

Today's Agenda

- Types of Memory


Computer Memory
Internal/Main/Primary
External/Secondary
RAM
ROM
HDD
SRAM
SSD
DRAM
Compact Disc
USB Flash Drive

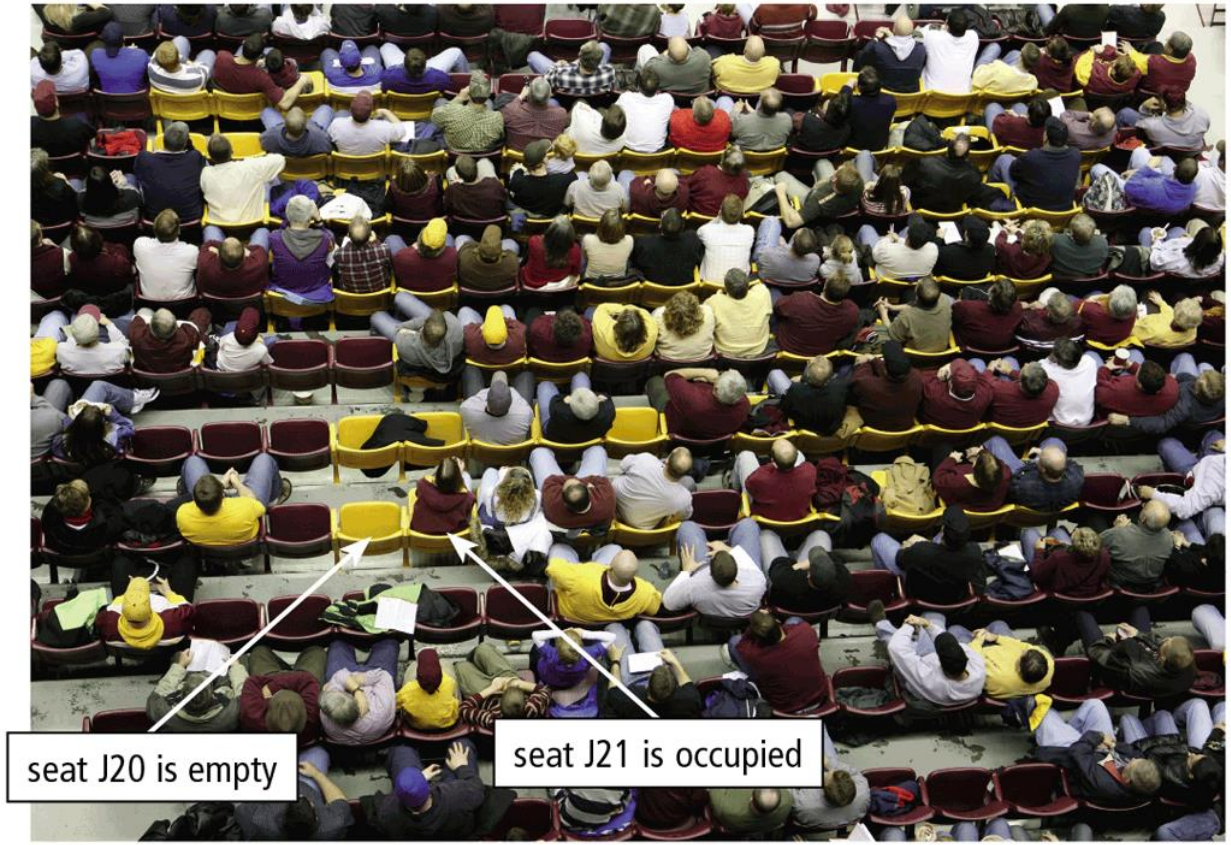
Memory

- Memory consists of electronic components that store three basic categories of items:



Memory

- Each location in memory has an address



Volatile vs Non-volatile Memory

Volatile memory

- Loses its contents when power is turned off
- Example includes **RAM and Cache**

Nonvolatile memory

- Does not lose contents when power is removed
- Examples include ROM, flash memory, and CMOS

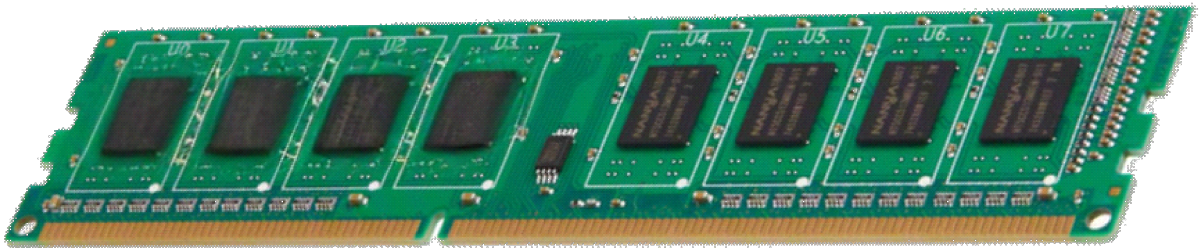
Volatile vs Non-volatile Memory

S.No	Volatile Memory	Non Volatile Memory
1	Volatile memory is the type of memory where data is lost when power is turned off	Non Volatile Memory is a type of memory where the data is not lost when a computer is switched off.
2	Data temporarily stored in volatile memory	Data permanently stored in non volatile memory
3	It is faster than non-volatile memory.	It is slower than volatile memory.
4	It has less storage capacity	It has more storage capacity than volatile memory
5	Data can be easily transferred	Data can not be easily transferred

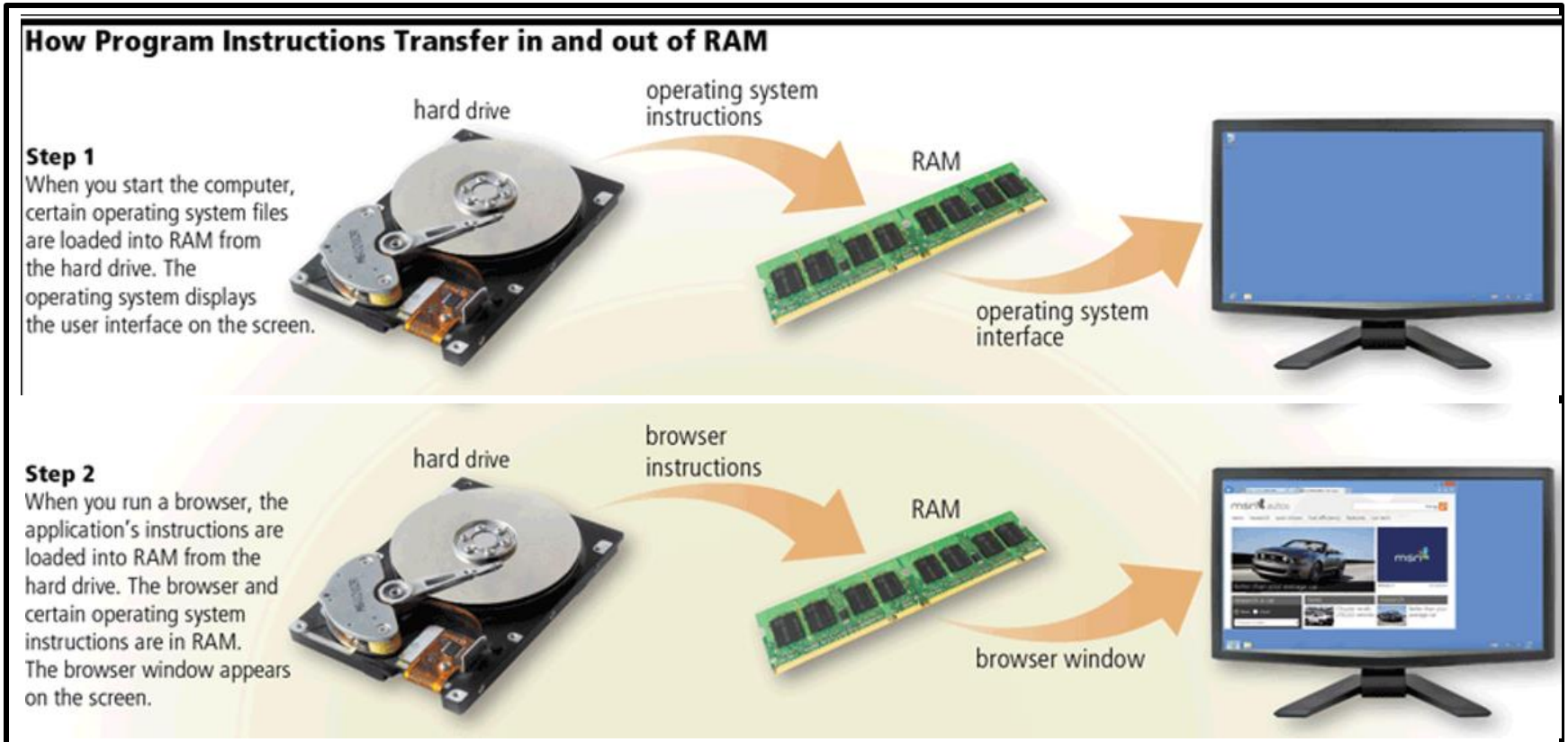
Volatile vs Non-volatile Memory

6	It is more costly per unit size.	It is less costly per unit size.
7	CPU has direct access to data.	CPU has no direct access to data.
8	Process can read and write	Process can only read.
9	It has a high impact on the system's performance.	It has a high impact on a system's storage capacity.
10	Data and programs that are currently fetch by CPU are stored in Volatile memory	Any kind of data and programs are stored in Non Volatile memory
11	Example: RAM and Cache Memory	Example: ROM and HDD

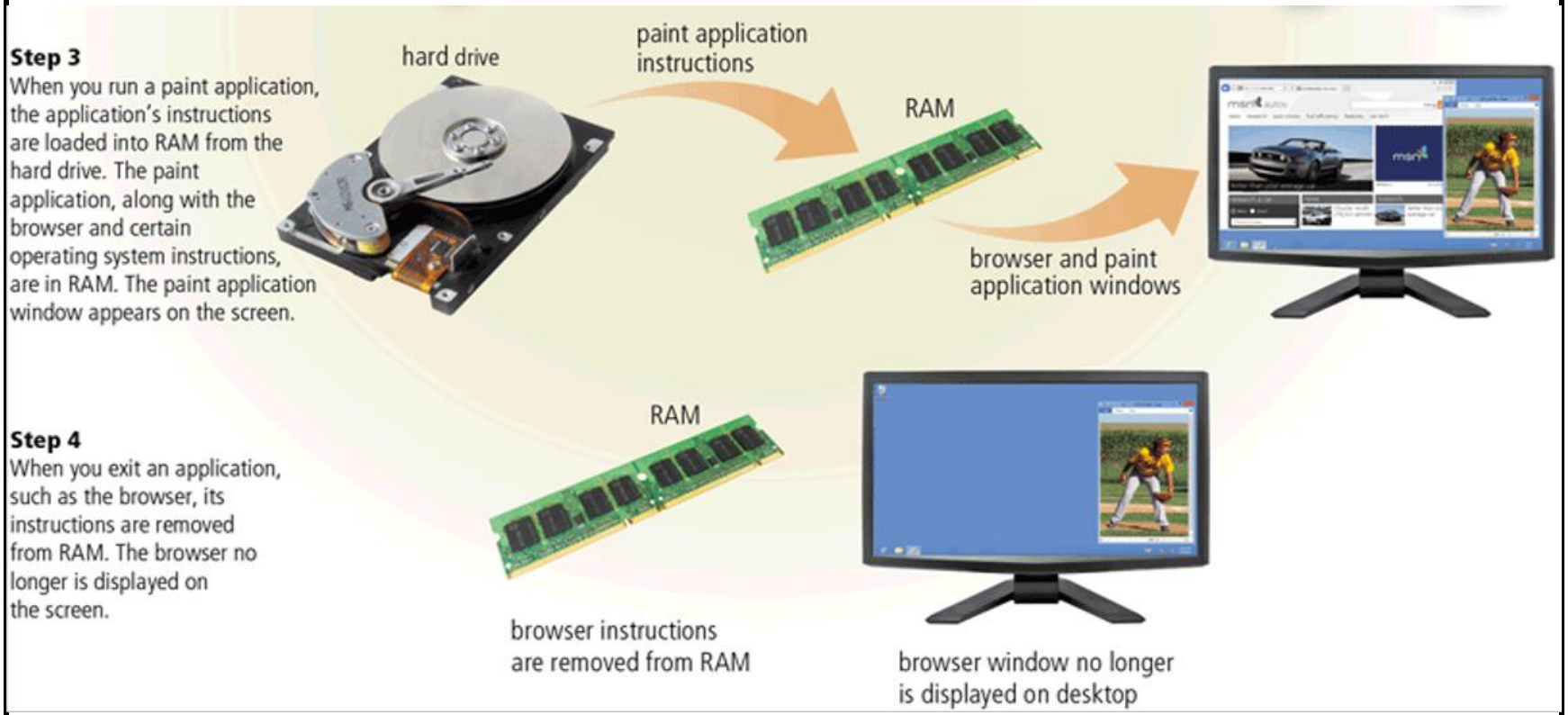

SAMSUNG MZVL



Memory

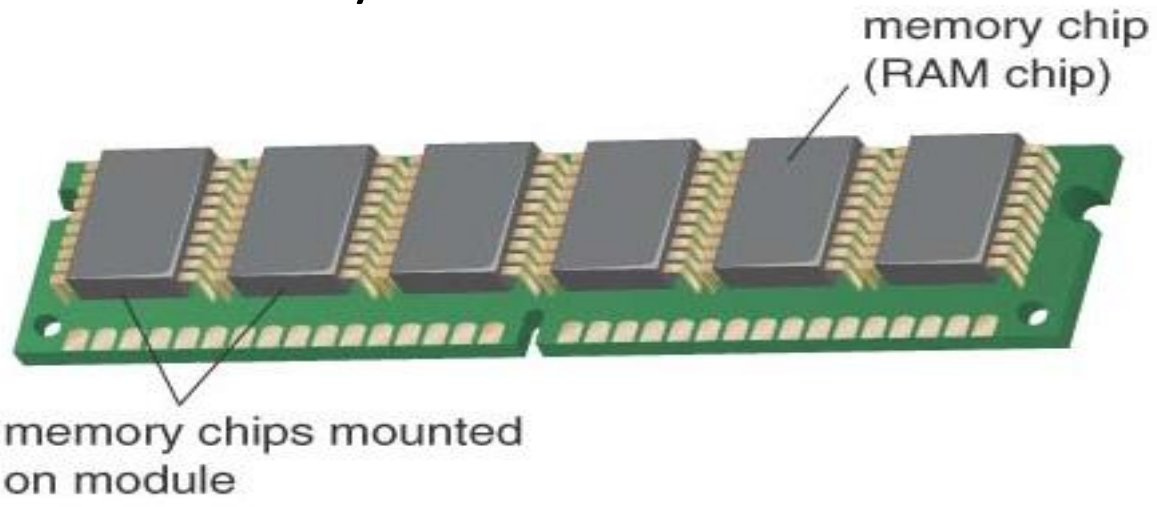

How Program Instructions Transfer in and out of RAM
Step 1
When you start the computer, certain operating system files are loaded into RAM from the hard drive. The operating system displays the user interface on the screen.
hard drive
operating system instructions
RAM
operating system interface
Step 2
When you run a browser, the application's instructions are loaded into RAM from the hard drive. The browser and certain operating system instructions are in RAM. The browser window appears on the screen.
hard drive
browser instructions
RAM
browser window

Memory


Step 3
When you run a paint application, the application's instructions are loaded into RAM from the hard drive. The paint application, along with the browser and certain operating system instructions, are in RAM. The paint application window appears on the screen.
hard drive
paint application instructions
RAM
browser and paint application windows
Step 4
When you exit an application, such as the browser, its instructions are removed from RAM. The browser no longer is displayed on the screen.
RAM
browser instructions are removed from RAM
browser window no longer is displayed on desktop

RAM: Random Access Memory

- RAM is a temporary memory bank where your computer stores data it needs to retrieve quickly.
- Most fundamental elements of computing
- RAM is volatile memory that temporarily stores the files you are working on.
- More RAM results in a faster system



ROM: Read Only Memory

- **Read-only memory** (**ROM**) refers to memory chips storing permanent data and instructions
- ROM is **non-volatile memory** that permanently stores instructions for your computer.
- Holds the computer boot directions



Types of RAM

- Types of RAM
  - Static
  - Dynamic
- Static RAM is faster and more expensive, used for CPU cache, and doesn't need refreshing because it uses transistors.
- Dynamic RAM is slower, cheaper, and used for main computer memory, but requires a constant refresh cycle to retain data because it uses capacitors.
- Both are volatile types of RAM, meaning they lose data when power is removed.

Types of ROM

- Types of ROM
  - Masked ROM (MROM): Data is permanently programmed during the manufacturing process and cannot be changed or erased.
  - Programmable ROM: Data can only be written by user once
  - Erasable Programmable ROM: Data can be erased using UV rays
  - Electrically Erasable Programmable ROM: Data can be erased using electric charge

CMOS

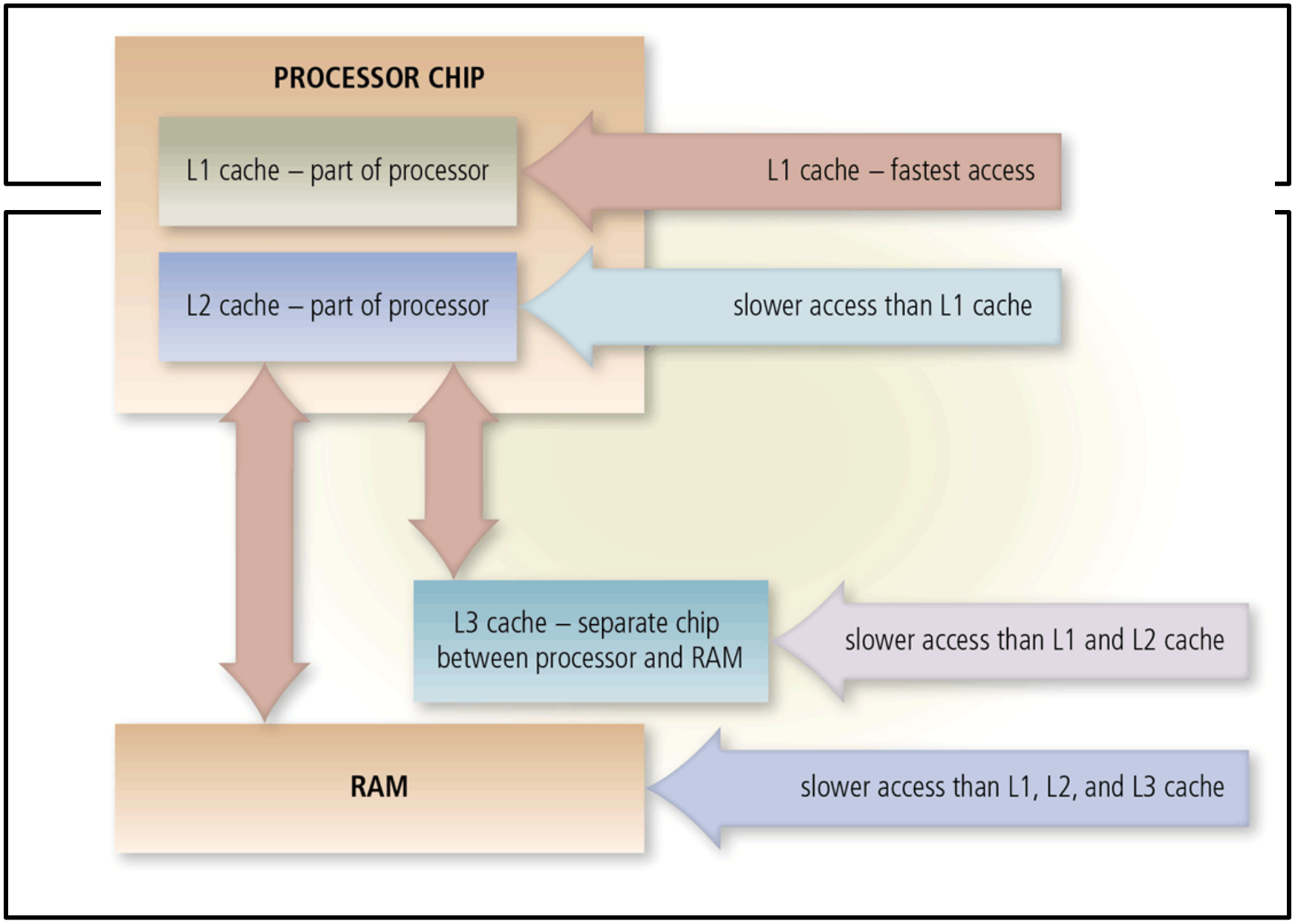
- Complementary Metal-Oxide-Semiconductor
- It refers to a small amount of memory on the motherboard that stores BIOS settings like the system time and date.
- It's a small low-power RAM
- Volatile, but is kept powered by a small battery on the motherboard to retain settings when the computer is off.

BIOS

- BIOS stands for Basic Input/Output System.
- It is a firmware that is the first program to run when a computer is turned on, responsible for initializing hardware and loading the operating system.
- BIOS is powered by a CMOS battery, which is why the computer remembers its settings even when turned off.
- BIOS checks and initializes system hardware components like the CPU, memory, and hard drive. It then finds and loads the operating system from the storage device into the main memory.
- It's a ROM, stored on an EEPROM
- Non-volatile, meaning it keeps its data even when the power is off, so the computer can always boot.

Cache

- A cache is a high-speed data storage layer, often in RAM, that stores frequently used instructions and data

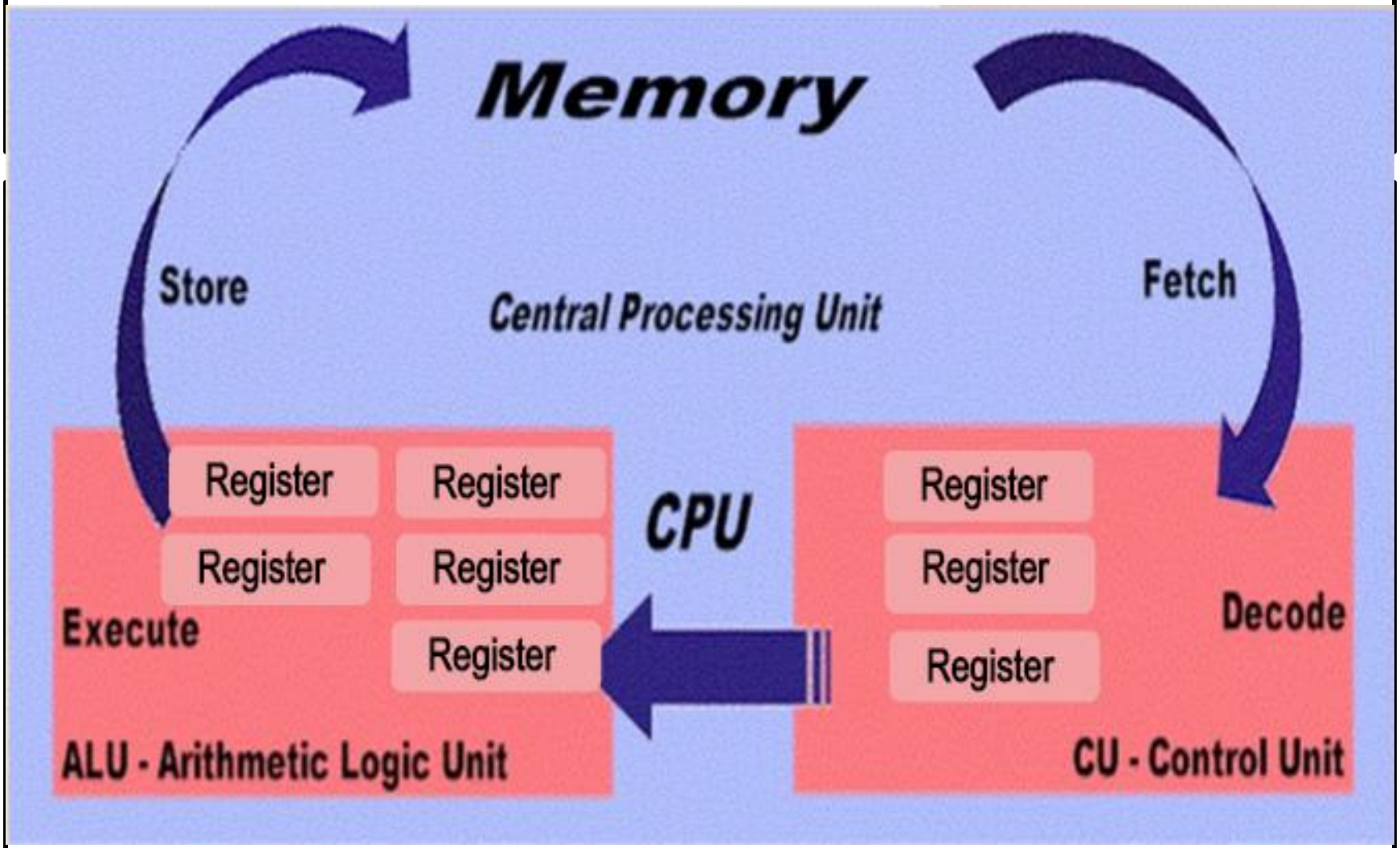

PROCESSOR CHIP
L1 cache – part of processor
L1 cache – fastest access
L2 cache – part of processor
slower access than L1 cache
L3 cache – separate chip between processor and RAM
slower access than L1 and L2 cache
RAM
slower access than L1, L2, and L3 cache

Cache

- L1 cache: The fastest, Smallest, built directly into each processor chip.
- L2 cache: Slower than L1 but faster than L3, Larger than L1, Can be on the CPU chip or on a separate chip connected via a high-speed bus
- L3 cache: The slowest, but still significantly faster than RAM. Size: largest, located off-core but on the processor chip, shared by all cores.

Registers

- A processor contains high speed and small sized locations, called registers
- Temporarily hold data and instruction during processing
- Size of register is 2 to 4 bytes
- Part of processor, but not part of main memory or storage devices


Memory
Store
Central Processing Unit
Fetch
Register
Register
Register
Register
Register
Register
CPU
Register
Register
Register
Execute
Decode
ALU - Arithmetic Logic Unit
CU - Control Unit

Types of CPU Registers

Register Symbol	Register Name	Function
DR	Data Register	Holds Memory Operand
AR	Address Register	Holds address for memory
AC	Accumulator	Processor Register
IR	Instruction Register	Holds Instruction Codes
PC	Program Counter	Holds Address of instruction
TR	Temporary Register	Holds temporary data
INPR	Input Register	Holds input character
OUTR	Output Register	Holds output character

Types of CPU Registers

- Program Counter (PC): The Program Counter holds memory address of the next instruction to be fetched and executed.

- Instruction Register (IR): The Instruction Register holds the currently fetched instruction being executed.

Types of CPU Registers

- Accumulator (ACC): The Accumulator is a general-purpose register used for arithmetic and logical operations. It stores intermediate results during calculations.
- General-Purpose Registers (R0, R1, R2…): These registers are used to store data during calculations and data manipulation. They can be accessed and utilized by the programmer for various purposes.

Types of CPU Registers

- Address Registers (AR): Address Registers store memory addresses for data access or for transferring data between different memory locations.
- Stack Pointer (SP): The Stack Pointer points to the top of the stack, which is a region of memory used for temporary storage during function calls and other operations.

Types of CPU Registers

- Data Registers (DR): These registers store data fetched from memory or obtained from input/output operations.
- Status Register/Flags Register (SR): The Status Register or Flags Register contains individual bits that indicate the outcome of operations, such as carry, overflow, zero result, and others. These flags help in making decisions and controlling program flow based on the results of previous operations.

Types of CPU Registers

- Control Registers (CR): Control Registers manage various control settings and parameters related to the CPU's operation, such as interrupt handling, memory management, and system configuration.

Access time

- Access time is the amount of time it takes the processor to read from memory
- Measured in nanoseconds



Reference/Self Study

- Ch-6